

HU Extension Assignment 02 E89 Advanced Deep Learning

Posted on: 09/11/2026

Due by 1:00 PM EST on Saturday, 09/19/2026

It is our advice that you work with Anaconda and use Conda to create virtual environments. If you know what you are doing, choose your environment as you please. The virtual environment for this course should have Python 3.10 or higher and PyTorch 2.9 or higher. Most code in our examples will work with lower versions of PyTorch, though.

You should have a local installation of PyTorch. We will not penalize you if you do not.

Problem 1. Create a conda virtual environment with Python 3.10 or higher on your local machine. Install PyTorch and Jupyter in that environment. In one or several cells of your submission Jupyter notebook test whether you have a cuda device. You will not be penalized if you do not have a GPU card on your machine. We just want you to find the way to make the test and perform the test. Perform such test in Google Colab as well. (30%)

Problem 2. Create a Python function of two arguments, x and y

$$\text{loss}(x, y) = \frac{x}{y + x^2} + e^{-yx}$$

Calculate the gradients of that function at points $(x, y) = (0, 1)$ and $(x, y) = (2, -2)$. Perform the calculations using Calculus and Pytorch Autograd. Compare two sets of results. (30%)

Problem 3. Consider the code in the notebook `3_optimizers.ipynb`, provided within Week 2 materials. Extend the dataset to 100 points by creating a random cloud of data points. For example, you could start with an equally spaced set of points between -20 and 45°C. Calculate Fahrenheit values using exact transformation. To both values add Gaussian noise of standard deviation 3°C. Transform your data into two datasets 70% for training and 30% for the validation data. Treat Celsius values as features and Fahrenheit values as targets. Please shuffle your data. Construct corresponding dataloaders with `batch = 10`. Use Adam as the optimizer and Mean Square Error as the loss function. Run your training for 3000 epochs. Trace and eventually plot training loss and the validation loss. Please present the scatter plot of your data and the prediction curve. You have to read API documentation to learn how to use Data Loaders. (40%)

Formatting Requirements:

Please name your notebook as `e89-LastName-FirstName-HW02.ipynb`

Solution of this assignment should be submitted as one Jupyter notebook (ipynb file), as well as an HTML image of that notebook. To generate the HTML image of you notebook, use the following command:

```
$ jupyter nbconvert e89-LastName-FirstName-HW02.ipynb --to html  
Or  
$ jupyter nbconvert --to html e89-LastName-FirstName-HW02.ipynb
```

Please provide meaningful descriptions of all actions in all cells of your notebook. Include those descriptions either as Python comments (#) or as the contents of the adjacent markdown cells.

If your notebook(s) contain(s) excessively long outputs, please copy a meaningful and illustrative number of initial and/or final lines and paste those in a markdown (comment) cell. Then, delete the long output(s). Thank you.